Exercise 5.8

Is C_2 uniquely decodeable?

Solution:

Yes, because we can determine all the position where 101 is from where 0s are. Then, we can determine that the remaining 1s are coded from 1s. The codewords can be uniquely decomposed and decoded.

Exercise 5.14

Prove the result stated above, that for any set of codeword lengths $\{l_i\}$ satisfying the Kraft inequality, there is a prefix code having those lengths.

Solution:

Sort l_i in descending order, and fill them into the budget diagram from top to bottom.
When the Kraft inequality is satisfied, there will always be a fit without any overlaps i.e. a prefix code.

Exercise 5.16

Prove that there is no better symbol code for a source than the Huffman code.

Solution:

Sibling lemma:
In an optimal code, if $p_i < p_j$ then $l_i \geq l_j$ since otherwise we can swap p for i and j to strictly decrease L.
Also, two longest codewords must be equal length (otherwise truncate) and can be taken to be siblings (differ by only last bit).
These prove that two least-probable symbols can be constructed to be siblings at max depth.

Now, we prove that Huffman coding is optimal by induction on n (the number of symbols):

(i) Base: Optimal $L = 1$, which is what Huffman constructs.

(ii) Inductive step:
Assume Huffman coding is optimal for $n - 1$ symbols. Let a, b be the least probable symbols of n symbols.
Huffman merges a, b into a new symbol and applies the algorithm to the remaining $n - 1$ symbols (optimal by hypothesis with length L'), then

splits the new symbol into a and b. This means Huffman's length is $L_H = L' + p_a + p_b$.
Now, assume $L_{opt} \leq L_H$. By the lemma we can assume a, b are siblings and by merging we get a symbol code of $L_{opt} - p_a - p_b$. Since L' optimal, $L_{opt} - p_a - p_b \geq L'$.
Thus $L_{opt} \geq L' + p_a + p_b = L_H \geq L_opt$ so $L_H = L_{opt}$. Hence Huffman coding optimal for n symbols.

Exercise 5.19

Is the code $\{00, 11, 0101, 111, 1010, 100100, 0110\}$ uniquely decodeable?

Solution:

No, since '111111' can be decoded as '11/11/11' or '111/111'.

Exercise 5.20

Is the ternary code $\{00, 012, 0110, 0112, 100, 201, 212, 22\}$ uniquely decodeable?

Solution:

This is a prefix code so yes, it is uniquely decodable.

Exercise 5.21

Make Huffman codes for X^2, X^3 and X^4 where $\mathcal{A}_X = \{0, 1\}$ and $\mathcal{P}_X = \{0.9, 0.1\}$. Compute their expected lengths and compare them with the entropies $H(X^2)$, $H(X^3)$ and $H(X^4)$. Repeat this exercise for X^2 and X^4 where $\mathcal{P}_X = \{0.6, 0.4\}$.

Solution:

X^2

00	0.81	0
01	0.09	10
10	0.09	110
11	0.01	111

$L(C, X) = 0.81 \times 1 + 0.09 \times 2 + 0.1 = 1.29$
$H(X^2) = 0.9380$

X^3

000	0.729	0
001	0.081	100
010	0.081	101
100	0.081	110
011	0.009	11100
101	0.009	11101
110	0.009	11110
111	0.001	11111

We get $L(C, X) = 1.598$, $H(X^3) = 1.407$

X^4

0000	0.6561	0
0001	0.0729	100
0010	0.0729	101
0100	0.0729	110
1000	0.0729	1110
0011	0.0081	111000
0101	0.0081	111001
0110	0.0081	1111010
1001	0.0081	1111011
1010	0.0081	111110
1100	0.0081	1111110
0111	0.0009	11111100
1011	0.0009	11111101
1101	0.0009	111111110
1110	0.0009	1111111110
1111	0.0001	1111111111

We get $L(C, X) = 1.9702$, $H(X^4) = 1.876$

Now, for $\mathcal{P}_X = \{0.6, 0.4\}$, $H(X) = 0.9710$ so
$H(X^2) = 1.9420$ and $H(X^4) = 3.8840$.
With a similar procedure as above, we obtain
$L(C, X) = 2$ and $L(C, X) = 3.9248$ respectively.

Exercise 5.22

Find a probability distribution $\{p_1, p_2, p_3, p_4\}$ such that there are *two* optimal codes that assign different lengths $\{l_i\}$ to the four symbols.

Solution:

$\{\frac{1}{3}, \frac{1}{3}, \frac{1}{6}, \frac{1}{6}\}$ is one such distribution.
$l = \{00, 00, 01, 11\}$ and $l = \{0, 10, 110, 111\}$ have both $L = 2$.

Exercise 5.23

(Continuation of exercise 5.22.) Assume that the four probabilities $\{p_1, p_2, p_3, p_4\}$ are ordered such that $p_1 \geq p_2 \geq p_3 \geq p_4 \geq 0$. Let Q be the set of all probability vectors $\mathbf{p}$ such that there are *two* optimal codes with different lengths. Give a complete description of Q. Find three probability vectors $\mathbf{q}^{(1)}, \mathbf{q}^{(2)}, \mathbf{q}^{(3)}$, which are the convex hull of Q, i.e., such that any $\mathbf{p} \in Q$ can be written as

$$\mathbf{p} = \mu_1 \mathbf{q}^{(1)} + \mu_2 \mathbf{q}^{(2)} + \mu_3 \mathbf{q}^{(3)}, \tag{5.29}$$

where $\{\mu_i\}$ are positive.

Solution:

For there to be two optimal codes with different lengths, since there are 3 merge taken, there must be 2 different ways to merge in the second merge (For first merge, it just makes it symmetric, for last, there is only 1 possible merge).
For that, p_1 must be equal to $p_3 + p_4$. Together with the other conditions,

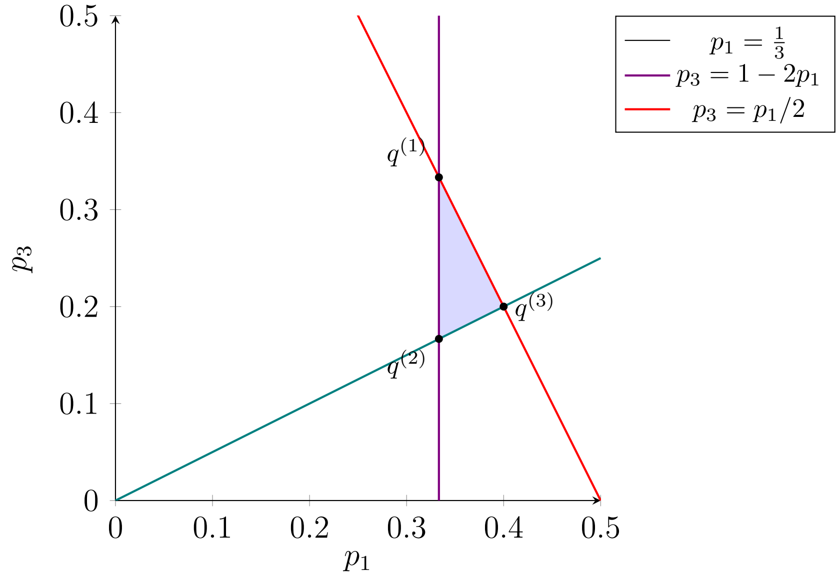
$$\begin{cases} p_1 = p_3 + p_4 \\ p_1 + p_2 + p_3 + p_4 = 1 \\ p_1 \geq p_2 \geq p_3 \geq p_4 \geq 0 \end{cases}$$

$\iff$

$$\begin{cases} p_1 = p_3 + p_4 \\ 2p_1 + p_2 = 1 \\ p_1 \geq p_2 \geq p_3 \geq p_4 \geq 0 \end{cases}$$

$\iff$

$$\begin{cases} p_1 \geq \frac{1}{3} \\ p_3 \leq 1 - 2p_1 \\ p_3 \geq \frac{p_1}{2} \end{cases}$$



The above area shows the possible p_1, p_3 pair (which uniquely determines p_2, p_4 from the conditions.)
Therefore, from linearity we get

$$
\begin{aligned}
q_1 &= (\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, 0) \\
q_2 &= (\frac{1}{3}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}) \\
q_3 &= (\frac{2}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5})
\end{aligned}
$$

Exercise

Solution: